Teardown Benchmark Report

Subject vehicle against Benchmark B and Benchmark C
Front axle, brake actuation and steering group

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Summary

Three 28-tonne rigid trucks were stripped to component level and the front axle, brake actuation and steering groups measured, weighed and costed on a common basis. The Subject vehicle front axle group is 12.6 kg heavier than the lightest benchmark, almost all of it in the forged beam, while its brake actuation group carries a 4.3 kg penalty arising from bracket design rather than from the chamber itself. Nine value-engineering opportunities are identified, of which four are assessed as implementable within the current platform without re-homologation.

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1 Scope and method

This report covers the front axle assembly, service brake actuation, hub and drum arrangement, front suspension bracketry and steering linkage of three 28-tonne rigid trucks. The scope excludes the rear bogie, the braking control electronics and the air supply upstream of the front reservoir, which are covered separately in CVOEM/VAVE/TD-BM/2024/012.

1.1 Teardown protocol

Each vehicle was received with between 1,800 km and 3,400 km accumulated and was stripped over four working days. Components were cleaned, individually weighed on a 0.01 kg resolution platform scale and dimensionally recorded. Fasteners were counted and classified by grade and coating. Where a component could not be separated without destroying it, mass was taken as an assembly and is shown as such in the tables.

1.2 Costing basis

All costs are fully absorbed piece costs at an assumed annual volume of 18,000 units, expressed in Indian rupees at October 2024 input prices. Material rates are taken from CVOEM/VAVE/MC-IX/2025/003. Tooling is amortised over four years on a straight-line basis. Competitor costs are estimated from measured mass, observed process route and published supplier rates; they carry an estimating tolerance of approximately 12 per cent and should not be read as procurement intelligence.

Method note. Competitor piece costs in this report are engineering estimates derived from teardown evidence, not quoted or contracted prices. They are adequate for ranking design alternatives and are not adequate for commercial negotiation.

2 Vehicles and build standard

Attribute	Subject vehicle	Benchmark B	Benchmark C
GVW (t)	28.0	28.0	28.0
Axle configuration	6x4 rigid	6x4 rigid	6x4 rigid
Front axle rating (t)	7.5	7.5	7.2
Front axle type	Forged I-beam, reverse Elliot	Forged I-beam, reverse Elliot	Forged I-beam, reverse Elliot
Service brake	S-cam drum, 410 x 180	S-cam drum, 410 x 180	S-cam drum, 410 x 160
Chamber size	Type 24	Type 24	Type 20
Front suspension	Parabolic, 3 leaf	Parabolic, 3 leaf	Parabolic, 2 leaf
Odometer at teardown (km)	2,140	3,380	1,860

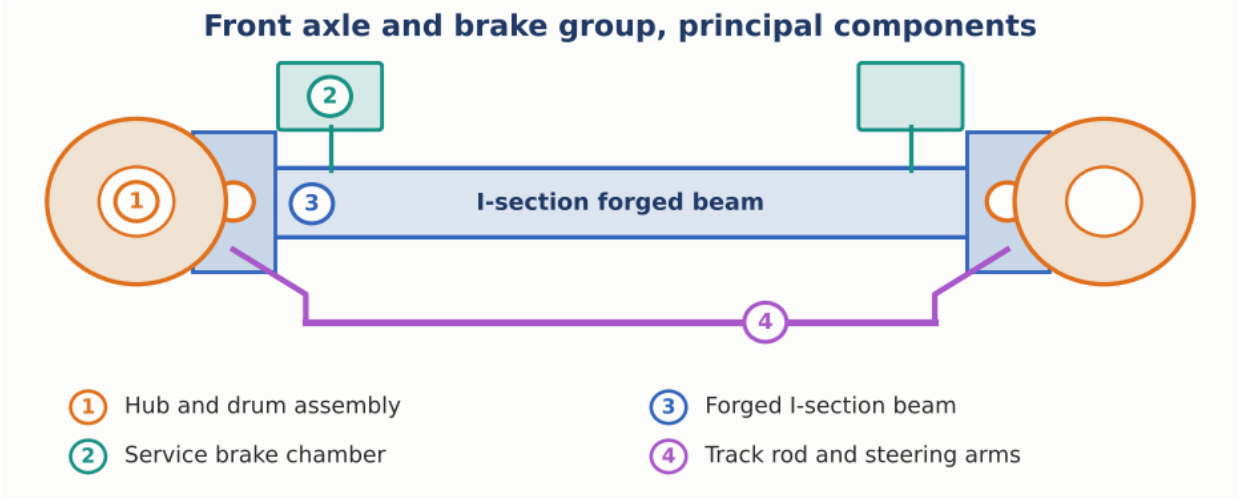


Figure 1 - Front axle and brake group, principal components. Numbering is used throughout sections 3 to 7.

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3 Front axle beam

3.1 Mass and section

The subject-vehicle beam is the heaviest of the three at 182.4 kg against 169.8 kg for the Benchmark C unit, a difference of 12.6 kg or 7.4 per cent. The difference is not distributed along the beam. Measured at the spring seat, all three beams carry a web of between 19.4 mm and 20.1 mm; at the mid-span the subject-vehicle web is 18.8 mm against 15.9 mm on the Benchmark C beam.

This is consistent with a beam designed to a uniform section for forging simplicity rather than to the bending moment distribution. The mid-span of a reverse Elliot beam carries substantially less bending moment than the spring seats under both vertical and braking load cases.

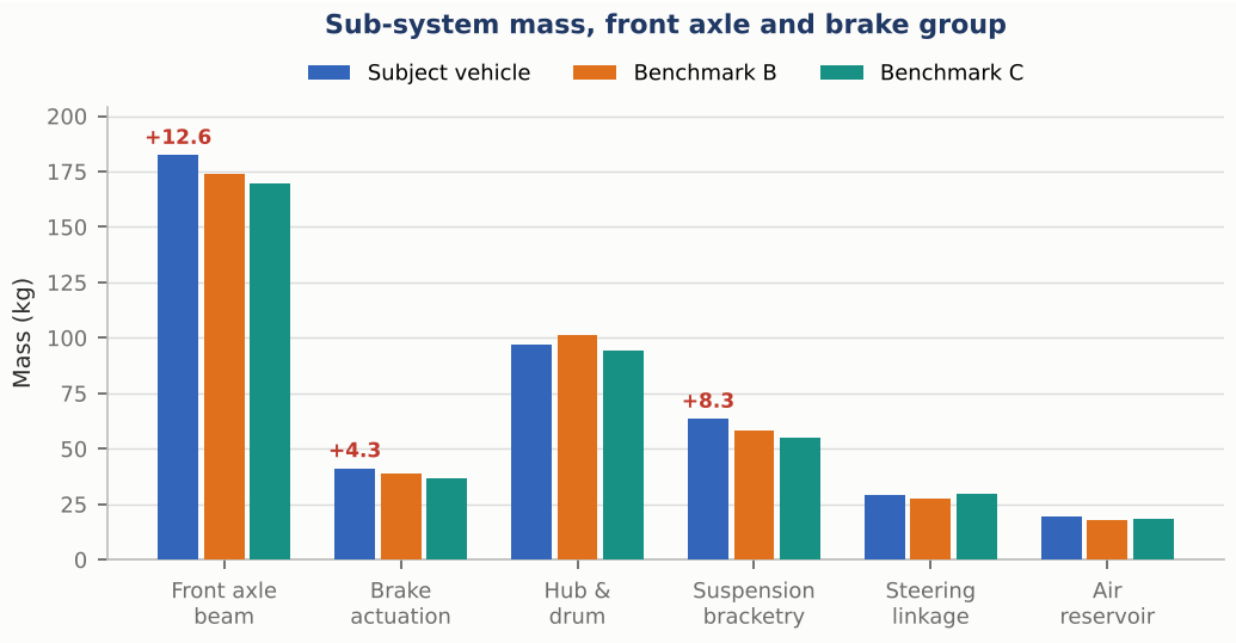


Figure 2 - Sub-system mass comparison. Annotations show the subject-vehicle position against the lightest benchmark; positive values are a mass penalty.

Finding 3.1. The front axle beam carries an estimated 9.5 kg of material that is not required by either the vertical or the braking load case. A tapered mid-span section, already used by one benchmark, would recover most of it without a change to forging equipment.

3.2 Material and process

Parameter	Subject vehicle	Benchmark B	Benchmark C
Material	EN 8D equivalent	EN 8D equivalent	Micro-alloyed 38MnVS6
Forging route	Closed die, 3 blow	Closed die, 3 blow	Closed die, 2 blow
Heat treatment	Quench and temper	Quench and temper	Controlled cooling from forging heat
Measured hardness (BHN)	262	255	248
Mass (kg)	182.4	174.1	169.8
Estimated cost (INR)	14,820	14,100	12,640

The Benchmark C beam uses a micro-alloyed steel cooled directly from forging heat, removing the separate quench and temper operation entirely. The measured hardness is 14 BHN lower but remains within the range normally specified for this application. The process saving is estimated at INR 1,240 per beam, of which approximately INR 860 is energy and handling in heat treatment.

Finding 3.2. Eliminating the separate quench and temper operation by moving to a micro-alloyed grade is the single largest cost opportunity identified in this teardown, at an estimated INR 1,240 per vehicle. It requires a fatigue re-validation and a supplier process capability study, and is therefore classified as a platform-level change rather than a running change.

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4 Brake actuation group

4.1 Chamber and bracket

The subject-vehicle brake actuation group is 4.3 kg heavier than the Benchmark C group. The chambers themselves differ by only 0.9 kg, reflecting the Type 24 against Type 20 selection. The remaining 3.4 kg sits in the mounting bracket: the subject-vehicle bracket is a six-web fabricated assembly of 6 mm plate, while the benchmark uses a 5 mm pressing with four webs and a local doubler at the highest-stress corner.

Component	Subject (kg)	Benchmark B (kg)	Benchmark C (kg)	Comment
Brake chamber, each	5.42	5.40	4.97	Type 24 against Type 20
Chamber bracket, each	3.86	3.11	2.44	Principal difference
Slack adjuster, each	2.18	2.20	2.09	Automatic, all three
S-cam shaft, each	4.90	4.78	4.81	Within tolerance
Camshaft bracket, each	1.84	1.61	1.52	Secondary difference
Air piping and fittings	2.42	2.18	2.61	Routing difference
Group total	41.2	38.6	36.9	

4.2 Bracket design comparison

The benchmark bracket achieves its mass reduction by accepting a higher local stress at one corner and addressing it with a doubler of 3 mm plate welded on the outer face, rather than by thickening the whole part. The approach is well established and is used elsewhere on the subject vehicle's chassis, notably at the spring hanger. It has not been applied to the brake chamber bracket.

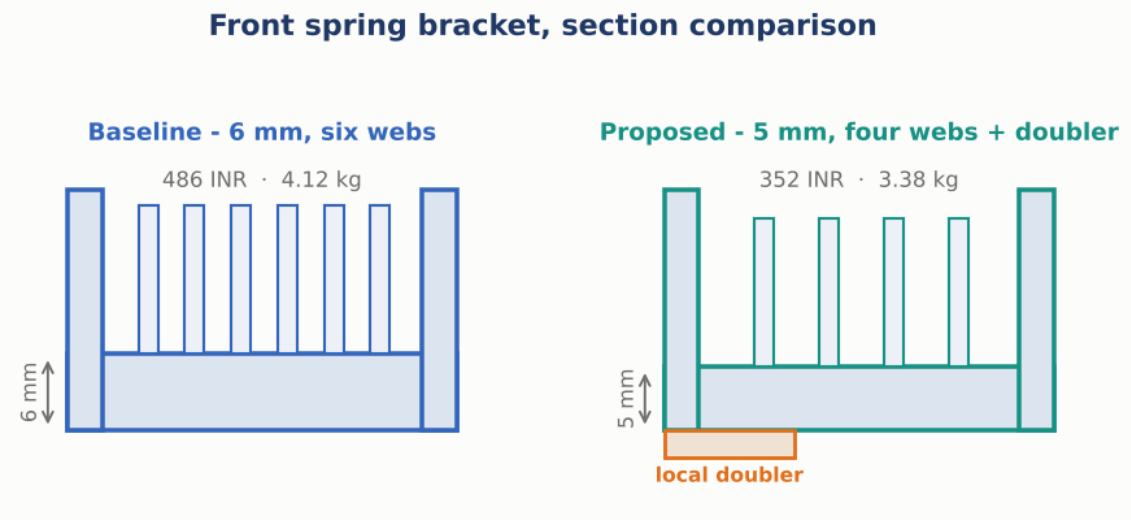


Figure 3 - Section comparison of the baseline fabricated bracket against the benchmark approach of a thinner section with a local doubler.

Finding 4.1. A 5 mm bracket with a local doubler would save 1.42 kg per side, 2.84 kg per vehicle, at an estimated piece-cost saving of INR 134. The doubler approach is already validated on this chassis at the spring hanger, which materially reduces the validation burden.

5 Hub, drum and bearing arrangement

The subject-vehicle hub and drum group is 4.4 kg lighter than the Benchmark B equivalent and 2.5 kg heavier than the Benchmark C group, which uses a 160 mm wide drum against 180 mm. The narrower drum reduces mass by 3.1 kg per side but raises lining temperature in the repeated-stop test by a measured 42 degrees Celsius. On the duty cycles relevant to Indian long-haul and tipper operation, the wider drum is retained as correct.

Finding 5.1. The 180 mm drum width is justified by thermal margin and is not a value-engineering opportunity. Any proposal to narrow the front drum should be closed with reference to this finding.

5.1 Bearing arrangement

All three vehicles use a taper roller pair with an oil-lubricated hub. The subject-vehicle arrangement uses a pressed steel cap with an O-ring seal; both benchmarks use a moulded polymer cap with an integral lip. The polymer cap is 0.21 kg lighter per side and removes a painting operation. Field evidence on cap retention is discussed in CVOEM/VAVE/WF-FA/2024/021.

6 Suspension bracketry

The front suspension bracketry shows an 8.3 kg penalty against the lightest benchmark. The front spring front bracket and the front spring rear bracket are separate forgings on the subject vehicle and on the Benchmark B vehicle; the Benchmark C uses a fabricated front bracket and a forged rear bracket, reflecting the different load paths at the two positions.

Finding 6.1. The front spring FRONT bracket and the front spring REAR bracket carry materially different load cases: the rear bracket reacts braking torque through the shackle while the front bracket does not. Proposals that treat the two as interchangeable should be rejected on this basis.

6.1 Process route and volume sensitivity

At the volumes relevant to this platform, the choice between fabrication, pressing and forging for the front bracket is finely balanced and depends principally on tooling amortisation. The analysis is set out in CVOEM/VAVE/SC-TC/2024/009; the break-even between fabrication and a progressive press tool falls at approximately 21,400 units per year.

7 Steering linkage

The steering linkage group is within 2.4 kg across all three vehicles and shows no significant opportunity. The subject-vehicle track rod uses a seamless tube of 38 mm outside diameter and 5 mm wall; both benchmarks use 36 mm by 5 mm. The difference reflects a longer rod and is appropriate.

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8 Comparative cost analysis

Group	Subject (INR)	Benchmark B (INR)	Benchmark C (INR)	Gap to best
Front axle beam	14,820	14,100	12,640	+2,180
Brake actuation	11,240	10,860	9,980	+1,260
Hub, drum, bearings	8,460	8,920	8,110	+350
Suspension bracketry	6,980	6,240	5,860	+1,120
Steering linkage	4,310	4,180	4,460	+130
Air reservoir and mounts	2,140	2,010	2,220	+130
Total	47,950	46,310	43,270	+4,680

The total gap of INR 4,680 per vehicle against the lowest-cost benchmark is concentrated in two groups: the forged beam and the brake actuation bracketry together account for INR 3,440, or 74 per cent of the gap. This supports a focused rather than a broad value-engineering effort.

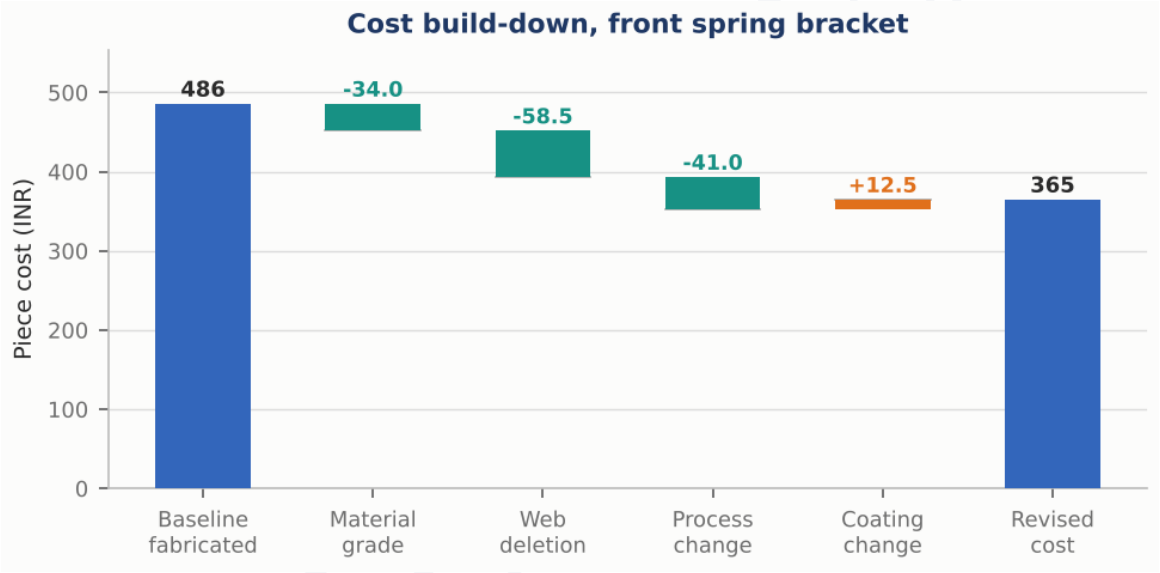


Figure 4 - Illustrative cost build-down for the front spring bracket, showing the contribution of each change and the cost added by the improved coating.

9 Value engineering opportunities

Ref	Opportunity	Saving (INR/veh)	Mass (kg/veh)	Classification
VE-15-01	Micro-alloyed beam, direct cooled	1,240	0.0	Platform change
VE-15-02	Tapered mid-span beam section	780	9.5	Platform change
VE-15-03	5 mm chamber bracket with doubler	268	2.84	Running change
VE-15-04	Polymer hub cap	96	0.42	Running change
VE-15-05	Camshaft bracket web deletion	74	0.64	Running change
VE-15-06	Common fastener family across group	112	0.08	Running change
VE-15-07	Air pipe routing simplification	148	0.24	Requires packaging review
VE-15-08	Fabricated front spring front bracket	420	1.90	Volume dependent
VE-15-09	Reservoir bracket commonisation	86	0.31	Requires packaging review

Finding 9.1. The four opportunities classified as running changes total INR 550 per vehicle and 3.98 kg, and require no re-homologation. At the platform volume of 18,000 units this is INR 9.9 million per year and is recommended for immediate release to the VAVE committee.

9.1 Opportunities considered and rejected

- Narrowing the front drum to 160 mm. Rejected on thermal margin; see finding 5.1. Mass benefit 6.2 kg per vehicle.
- Aluminium chamber bracket. Rejected. The equivalent change on the cab mounting bracket was reversed in FY24 following field fatigue cracking; see CVOEM/VAVE/WF-FA/2024/021 section 4.
- Commonising the front spring front and rear brackets. Rejected on load case; see finding 6.1.
- Two-leaf parabolic front spring. Deferred. The benchmark achieves this with a higher-rate interleaf pad; the ride consequence has not been assessed.

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10 Conclusions and recommendations

The Subject vehicle front axle and brake group is competitive in function and carries a cost gap of INR 4,680 and a mass gap of 12.6 kg against the best benchmark in this study. Seventy-four per cent of the cost gap is concentrated in the forged beam and the brake actuation bracketry.

- Release VE-15-03, VE-15-04, VE-15-05 and VE-15-06 to the VAVE committee as running changes, total INR 550 and 3.98 kg per vehicle.
- Open a platform-level study on the micro-alloyed direct-cooled beam, VE-15-01 and VE-15-02 taken together, with a fatigue validation plan.
- Record findings 5.1 and 6.1 in the idea register so that future proposals to narrow the front drum or commonise the spring brackets can be closed by reference rather than re-analysed.
- Re-assess VE-15-08 when the FY26 volume forecast is confirmed, against the break-even in CVOEM/VAVE/SC-TC/2024/009.

Recommendation. The committee is asked to approve the four running changes and to authorise a feasibility study on the beam. No decision is sought on the deferred items in this submission.

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